Supplementary Information

ITAI 1371 – Introduction to Machine Learning | Midterm Project

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Part B – AI Usage Disclosure

Did you use any AI tools?

Yes. I used Claude (Anthropic) and GitHub Copilot during parts of this project. The table below explains exactly where and how.

AI Tool	Sections Used In	Type of Help
Claude (Anthropic)	Report writing, code structure	Draft text & code templates
GitHub Copilot	Jupyter notebook cells	Code autocomplete suggestions
No AI used	Discussion, model selection, results analysis	Written & decided independently

Report Writing (Claude)

I used Claude to help write the first draft of the Introduction, Literature Review, Problem Definition, and Conclusion sections. I gave Claude my project proposal and dataset details, and it produced an initial draft. I then went through each paragraph, edited the language, added specific numbers from my actual results, and rewrote sections that did not accurately describe my work.

Code Help (GitHub Copilot)

While writing the Jupyter notebook, Copilot suggested autocomplete for repetitive code like the GridSearchCV setup and the ROC curve plot. I reviewed each suggestion, ran the code, checked the output, and modified parameters based on my own understanding of what the models were doing.

What I did myself (no AI)

The following were entirely my own decisions, not AI-generated:

- Choosing to use MarketEstimate / ListedPrice as the engagement proxy
- Deciding on the 75th percentile as the High Engagement threshold
- Selecting which four models to compare and why
- The entire Discussion section (Section 5 of the report)
- Interpreting confusion matrices and identifying the recall/imbalance issue

How I understood the AI-generated code

For every cell in the notebook that had AI-assisted code, I: (1) read the code line by line before running it, (2) added my own comments explaining what each part does, (3) ran the cell and verified the output made sense, and (4) could explain the logic if asked. I did not copy-paste code I did not understand.

Part C – Code Logic Diagram

The diagram below shows the flow of the code from start to finish. Each box is one major step, and arrows show the order of execution.

Step 1
INPUT DATA
cleaned_df.csv
22,681 listings | 14 columns | Zillow 2023

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Step 2
FEATURE ENGINEERING
Build Engagement Target
Engagement Score = MarketEstimate / ListedPrice
Label = 1 if top 25%, else 0

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Step 3
DATA PREPROCESSING
Clean & Prepare
Impute missing values (median) | Encode 'State'
Scale features | 80/20 train-test split

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Step 4
MODEL TRAINING
Train 4 Classifiers
Logistic Regression | Decision Tree
Random Forest | Gradient Boosting

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Step 5
HYPERPARAMETER TUNING
GridSearchCV on Random Forest
Search: n_estimators, max_depth, min_samples_split
3-fold cross-validation

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Step 6
EVALUATION
Measure Performance
Accuracy | Precision | Recall
F1 Score | ROC-AUC | Confusion Matrix

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Step 7
OUTPUT / RESULT
Best Model: Gradient Boosting
F1 Score = 0.403 | ROC-AUC = 0.806

Figure: Code workflow — from raw CSV input through feature engineering, preprocessing, model training, hyperparameter tuning, and evaluation to the final prediction output.